MANIPAL UNIVERSITY JAIPUR

School of Computer Science & Engineering

Department of Computer Science & Engineering

PROJECT SYNOPSIS

for Predictive Analytics

Project Title

“ReturnIQ – Smart Return Prediction for Online Retail”

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Course / Lab	Predictive Analytics
Semester	V
University	Manipal University Jaipur

SYNOPSIS

1. Project Title

ReturnIQ – Smart Return Prediction for Online Retail

2. Abstract

This project proposes the design and implementation of a predictive analytics system that estimates the likelihood of a purchased product being returned by a customer. The system will demonstrate the practical application of predictive analytics and machine learning concepts, including data cleaning, point-in-time feature engineering, chronological train/validation/test splitting, benchmarking of classification models, threshold optimization based on business cost, and model interpretability. Order and customer data will be processed to build a binary classification model that predicts whether a product line item will be returned, so that the platform can take proactive steps such as better sizing guidance or review prompts before the return happens. The project focuses on keeping the workflow simple to understand while covering the complete predictive analytics pipeline, from raw data to an explainable, cost-optimized model.

3. Problem Statement

E-commerce platforms lose a significant amount of revenue due to product returns caused by wrong sizing, mismatched expectations, or poor product fit. Most platforms only react after a return has already happened, which adds logistics and reverse-shipping costs. The objective of this project is to build a predictive system that can flag high-risk orders at checkout itself, using historical order and customer behaviour, so that the platform can intervene early and reduce avoidable returns.

4. Objectives

- To collect and clean raw order and customer data for analysis.
- To perform point-in-time feature engineering that avoids using information not available at prediction time (prevents target leakage).
- To split the data chronologically into training, validation, and test sets so the model is tested the same way it would be used in real life.
- To build and benchmark multiple classification models such as Logistic Regression, Random Forest, LightGBM, and CatBoost.
- To optimize the model's decision threshold using a cost-benefit approach so that the model saves the most money, not just gets the most accuracy.
- To explain model predictions using interpretability techniques such as SHAP summary and waterfall plots.

5. Proposed System Components

The system will be organized into the following modules: a Data Cleaning & Preparation module for raw order and customer data, a Feature Engineering module that builds point-in-time behavioural features, a Model Training & Benchmarking module that compares several classification algorithms, a Threshold

Optimization module that converts predicted probabilities into a business decision using a custom cost-benefit curve, and an Interpretability module that explains why the model flagged a given order as high-risk.

6. System Workflow

▼ Raw Order & Customer Data
▼ Data Cleaning & Temporal Ordering
▼ Point-in-Time Feature Engineering (prevents target leakage)
▼ Chronological Train / Validation / Test Split
▼ Benchmark Classification Models (Logistic Regression, Random Forest, LightGBM, CatBoost)
▼ Financial Threshold Optimization (custom cost-benefit curve)
★ Global & Local Interpretability (SHAP summary & waterfall plots)

7. Predictive Analytics Concepts to be Demonstrated

- Supervised binary classification (return vs. no return).
- Data cleaning and chronological ordering of records.
- Point-in-time feature engineering to avoid target leakage.
- Chronological train/validation/test splitting.
- Model benchmarking across Logistic Regression, Random Forest, and gradient-boosted models (LightGBM, CatBoost).
- Cost-benefit based decision threshold optimization.
- Model interpretability using SHAP (Shapley Additive Explanations).

8. Expected Outcome

The expected outcome is a working predictive model prototype that can estimate the probability of a product return for a given order. The project will demonstrate correct point-in-time feature engineering, a fair chronological evaluation, a cost-optimized decision threshold that reduces reverse-logistics expenses, and clear, explainable reasoning behind each prediction.

9. Student Details

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